Part IV
Calibration – Overview

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13
Towards a Guideline for SPM Calibration

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Abstract

In the course of the evolution of many high technologies such as microelectronics, micromechanics, and also biotechnology, the size of technical structures is being decreased continuously. In many technical applications, the feature size has already reached the submicron scale and often needs to be measured with an uncertainty in the nanometer range. Scanning probe microscopes (SPMs) are therefore increasingly used today as quantitative measurement instruments. Consequently, the demand for standardized calibration routines for these types of instruments rises. A number of transfer standards suited for SPMs have already been developed for this purpose and are commercially available. While detailed knowledge of the standards' properties is a prerequisite for their practical application, the calibration procedure itself also deserves careful consideration. Up to now, there is no generally accepted guideline on how to perform SPM calibrations. This chapter therefore discusses a possible calibration scheme and focuses on several critical aspects of SPM characterization, e. g., the determination of the static and dynamic physical properties of the instrument and its ultimate limits, the influence factors that need to be considered when plotting a scheme for the calibration of its scan axes, and the possible error sources that need to be checked to ensure that all uncertainty contributions are taken into account on calculation of the measurement uncertainty according to *Guide to the Expression of Uncertainty in Measurements* (GUM).

13.1
Introduction

Scanning probe microscopy (SPM) is increasingly used today not only in research and development but also in many fields of industrial fabrication and inspection. Growing importance is therefore attached to the *quantitative* information these instruments provide. Consequently, there is an increasing need to check the accu-

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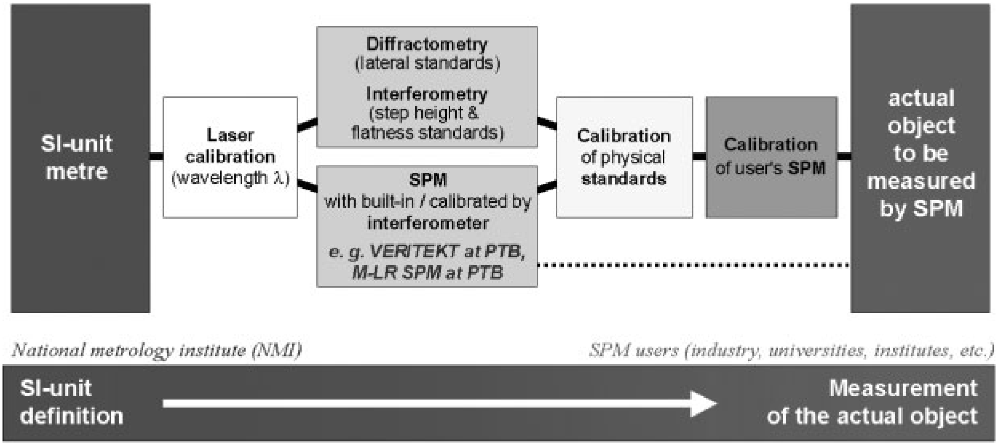

Fig. 13.1 Traceability chain for SPM.

racy of the SPM used in a similar way as it is already routinely practiced with coordinate-measuring instruments used for larger scale measurements and to make sure that those measurements are traceable to the SI unit (see Figure 13.1).

However, except for a few specially designed setups mainly at national metrology institutes, measurements made with these instruments usually lack traceability to the meter definition. In an attempt to fill this gap, several commercial companies and research institutes have developed a number of transfer standards in recent years that are especially suited for the needs of SPMs [1, 2]. They may be divided into those for the actual calibration and others for the verification of the instrument (Figure 13.2).

The first types of standards consist of regular structures of well-known geometry with calibrated dimension(s) in lateral and/or vertical direction and are thus meant to allow the calibration of SPMs, e. g., single axis x, y, or z, to look for rectangularity of the axes, etc.

Calibration	Verification of properties
• x- and y-axes • z-axis • Angle errors of axes (yaw, pitch, roll) • Rectangularity of the axes	• Influence of external mechanical and acoustic vibrations, electrical noise, air flow • Noise of the instrument • Thermal drift • Internal noise • Scanner guidance properties • Tip characteristics

The other types of standards or samples are needed to verify the behavior of the instrument at its location respectively in its environment such as its susceptibility toward ambient conditions (noise, drift) and to determine its internal properties

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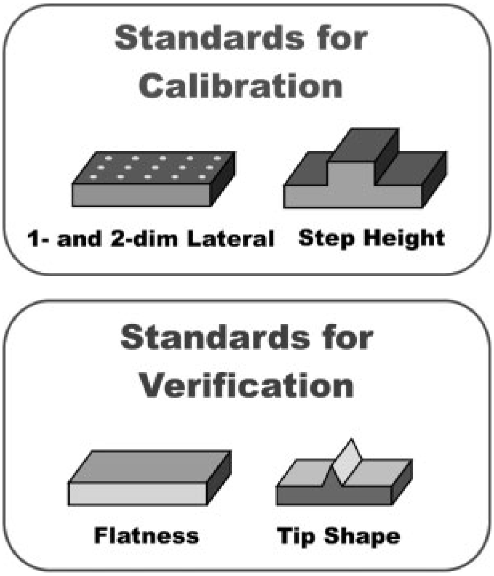

Fig. 13.2 Calibration of SPM and verification of properties.

such as warm-up characteristics and the shape of the tip used. While tip problems may be solved easily by changing the probe, most of the other properties can be measured, but many of them often cannot be adjusted by simple means. For example, the noise of the instrument may usually be recorded easily, while it typically takes a considerable effort to reduce the noise level, particularly when changes to the laboratory turn out to be inevitable.

A guideline should therefore help the user to determine the right calibration strategy with respect to his or her instrument type, to define an agreeable characterization/correction effort according to his or her actual needs and available resources, to perform the necessary calibration steps in the right way, and to draw the correct conclusions. Consequently, the aims of such a guideline for SPM are as follows:

- the definition of metrological terms, symbols, procedures, and standards
- minimal requirements for the calibration of an instrument
- criteria to determine the degree to which an instrument can be calibrated reasonably
- determination of the uncertainty of measurements
- the improvement of the comparability of measurements of geometrical properties

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13.2
General

13.2.1
Schematic Description of SPMs

Figure 13.3 shows the schematics of an SPM with the main components as used in the chapter.

The main components of SPMs are as follows (Figure 13.3):

x–y scanner	element to displace the probe (scanning probe microscope) or alternatively the sample (scanning sample microscope) usually in equidistant steps or continuously at constant speed
z actuator	element to keep the probe under constant conditions (usually at a constant value of the interaction used for control) while scanning the sample surface
Probe	fine tip for the investigation of the sample properties
Sample	object to be investigated by SPM
Feedback sensor	element that allows us to detect small changes of the tip or cantilever properties (i. e., of the interaction exploited)
x–y coarse positioning	elements to position and align the sample to bring the region of interest of the sample into the reach of the x–y scanner
z coarse approach	element to decrease the displacement between the sample and the probe to a value that allows the z actuator to bring the probe and the sample into interaction

The SPM can be understood as a small coordinate measurement machine in which the probe is moved in all three directions, but with smaller ranges. Therefore the probe and the scanning system have all degrees of freedom. In three dimensions, this amounts to 21 degrees (see Figure 13.4): 3 positioning errors (*xpx*, *ypy*, *zpz*), 6 straightness errors (*ytx*, *xty*, *ztx*, *zty*, *xtz*, *ytz*), 9 rotational errors (*xrx*, *yry*, *zrz*, *xry*, *yrx*, *zrx*, *zry*, *xrz*, *yrz*), and 3 squareness errors (*xwy*, *xwz*, *ywz*).

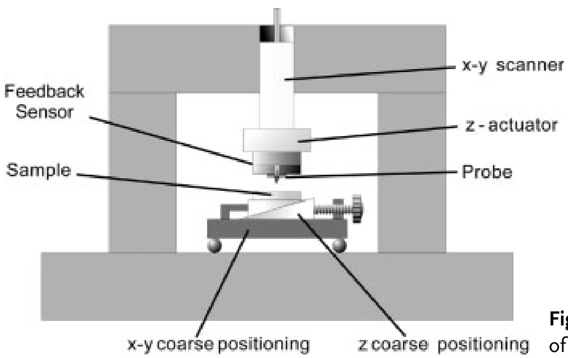

Fig. 13.3 Schematic description of the components of an SPM.

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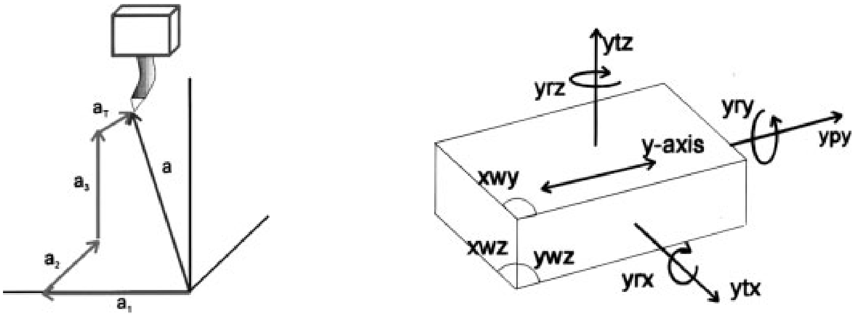

Fig. 13.4 (a) Vector **a** describing the movement of the tip in 3D. (b) Schematic description of the translation rotation axes and axes between the axes.

13.2.2 Metrological Classification of SPMs

If an SPM is not only expected to yield qualitative information but also intended for quantitative measurements, the traceability to the SI unit of length has to be realized. Before a calibration routine is plotted, it is therefore essential to consider how traceability is actually realized in the SPM system to be calibrated respectively how a traceability chain might be established.

Consequently, existing types of SPMs may – according to their properties – be divided into the following three categories with respect to the quality of their position measurements:

A Reference SPMs with integrated laser interferometers, traceability directly via the wavelength of the laser used is possible (often called "metrological SPM").

B SPMs with position monitoring by integrated position sensors, e. g., capacitive sensors, inductive sensors, strain gauges, encoders calibrated either by temporarily attaching laser interferometers to the scan system or by using high-quality physical transfer standards. This category comprises both SPMs with active position control (i. e., with a feedback circuit: so-called closed-loop SPMs) and without (open-loop SPMs with integrated position sensors for monitoring only).

C SPMs with positioning by simply using the voltage applied to the scanner (x & y) respectively with positions deduced from the voltage applied to the scanner (z), usually calibrated by using physical transfer standards.

As a consequence of this definition of categories, certain SPM systems may not be assigned to one category as a whole, as the equipment of each of its three axes might differ. In such a case, each axis needs to be treated individually.

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The category is a first important criterion for the planning of the calibration routine, as both the focus of the calibration procedure and the extent of the calibration effort largely depend on the metrological properties of the SPM. For instance, it is generally not advisable to purchase a set of very precise and thus rather expensive standards, e. g., of the uncertainty class 10^{-5} for the calibration of an SPM of category C. In such a case, a standard with an uncertainty of 10^{-3} will most likely prove sufficient.

13.2.3 Calibration Intervals

Another general classification describes the appropriate repetition rate of calibration. This depends both on the class of instrument and its stability and on the accuracy the user aims to maintain for the specific measurement purposes. As calibrations are usually quite time-consuming and thereby cost-intensive, a compromise is to be found between necessary calibration effort and tolerable uncertainty.

Especially when starting the operation of a new (or substantially modified) SPM system or changing its ambient conditions, frequent recalibrations are advisable in the initial phase. In order to find the best-suited strategy, experience needs to be gained for the particular SPM system to assess its behavior and deduce criteria for recalibration. The following schemes of calibrations (A) and measurements (B) may prove reasonable:

ABB... ABB...	instrument with a high medium-term stability: calibration necessary only in regular time intervals, e. g., weekly, monthly, annually
AB, AB, ...	instrument with sufficient short-time stability, but lacking satisfying long-time stability: calibration necessary before each measurement
ABA, ABA, ...	calibration before and after the measurement to exploit the maximum accuracy of the system; or the system shows poor stability–drift of the system's properties taken into account as closely as possible

13.3 Verification of Properties of Instrument, Tip and Environment

As a first step to develop a calibration procedure for SPM, all possible error sources that may influence the performance of the scan system need to be identified and assessed. Then, in a second step, suitable calibration routines have to be established. Examples for this procedure are given in the following.

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13.3.1 Ambient Conditions

Temperature changes, turbulences, vibrations, dust, and dirt as well as the presence of staff are ambient factors that deserve consideration before the SPM system is implemented. Sometimes the SPM is isolated from the environment by using an acoustic chamber. The drawback of such an acoustic chamber is that it might also act as thermal insulator. The heat sources and the time constants of the temperature change need to be determined to take appropriate counter measures. At least they should be taken into account when operation guidelines are plotted. Such a temperature dependence can be avoided – or at least minimized – in the course of the construction of an SPM system by taking great care of the measuring circle, i. e., the mechanical construction that links the probe with the sample. The basic idea is to construct the measuring circle in such a way that the probe's position relative to the sample does not change at all or at least not in all three directions. This is usually achieved by selecting materials with a low thermal expansion coefficient and designing the measurement circle to be as small as possible.

In order to avoid misleading conclusions and to reduce the calibration effort to the level actually required, the physical properties of the SPM system need to be taken into account when determining the best-suited strategy for the SPM's calibration. Especially the degree of reproducibility is decisive for the necessary and reasonable calibration effort.

13.3.2 Flatness Measurements and Signal Noise

Guidance errors and the noise in the measured signal are key figures of an SPM's principal mechanical and electronic limits. They set the ultimate limit for the necessary calibration effort reasonable for the particular SPM system. It is therefore advisable to determine these values before making a thorough calibration plan for the individual SPM system (see Table 13.1).

A first assessment of guidance errors is accomplished by measurement of a flatness standard, i. e., a sample with a well-defined reference area that has been calibrated by interference microscopy. Subtraction of these reference data from the actual SPM measurement data yields the out-of-z-plane movement across the lateral scan range of the measuring system (Figure 13.5).

By repeating such measurements, a distinction can be made between deviations from the z plane of temporary, initial nature (e. g., remaining temperature drift, creep of piezos), and permanent cross-talk in the z-direction by lateral movement in the x and y (e. g., scanner bow, detection/feedback errors).

In order to check the signal noise, measurements can be recorded without (i. e., lateral scanning disabled) and while scanning a small section, e. g., of a flatness standard [1]. The detected oscillations may be caused by mechanical or electronic influences.

Table 13.1 Instrument properties and their quantitative determination

Properties		
Drift	Vertical	Flatness standard or sample with flat regions. Measurements of a 2D grid with small height and looking for differences. Variation of environmental conditions. Opening of chamber. Switching on/off of instruments
	Lateral	Sample with straight edge or line with small step height parallel or perpendicular to the scan direction. Measurements of a 2D grid with small height and for differences
Guidance errors	Cross-talk *xtz*, *ytz*	Flatness standard
	Cross-talk *xty*, *ytx*	2D lateral standard or line respectively groove perpendicular to the fast and parallel to the slow scan direction.
z –noise	Static: without moving in the *x*- or y-direction	After stabilization of the instrument. Measurement on a flat sample region without structure while any movement is disabled. Additionally, variation of environment conditions, e. g., mechanical damping, acoustic vibrations, etc.
	Dynamic: scanning in the *x*- or *y*-direction	After stabilization of the instrument. Fast recording of two or more measurement lines on a flat sample region without any structure. The differences should give the dynamical noise.
x–*y* noise	Static	After stabilization of the instrument. Sample with straight edge or line with small step height parallel or perpendicular to the scan direction.

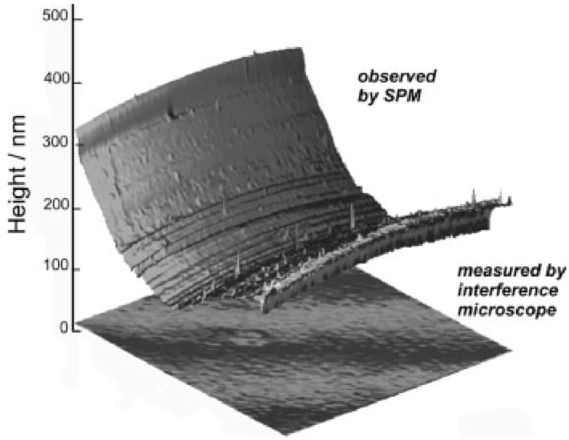

Fig. 13.5 Use of a flatness standard to determine out-of-*z*-plane movement of an SPM. *Lower plane*: surface of the flatness standard measured by interference microscopy as reference 100 μm × 100 μm, *upper plane*: the same region as observed by an SPM with a typical tube piezo scanner. By subtraction of the interferometrically measured plane, the flatness errors of the SPM can be determined, e. g., "scanner bow," drift, noise, and errors in the *z*-feedback control (here: spikes).

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Mechanically induced vibrations may consequently be reduced by passive means (e. g., a more massive support of the instrument, a better acoustic shielding/damping or, if possible, by alterations to the geometry of the instrument), or by active means (e. g., a closed-loop antivibration table).

Electronic oscillations may be caused by ambient interference or the feedback control settings. In the first case, a better electromagnetic shielding of the electronics might help, in the latter case a variation of the control parameters.

13.3.3
Repeatability and Noise

To illustrate repeatability and noise investigations, measurements of the stability and the noise of an STM equipped with capacitive transducers (SPM of category B) are discussed as a practical example. The same line was scanned several times without changing the y-position, first on a flat part of a sample, later in a structured region with holes 100 nm in depth and 1500 nm in width (see Figure 13.6).

Scanning lines under the same conditions on a structured region yields the results shown in Figure 13.7(a). To better illustrate the details, some successive lines are plotted in Figure 13.7(b). Here the sharp edges of the structure can be used to guide the eye. The fine structures at the tops and at the bottoms look very similar. The difference between the profiles obtained in two successive lines is shown in Figure 13.7(c). The maximum deviations are now in the range of ±5 nm. The positions at which these large differences occur clearly

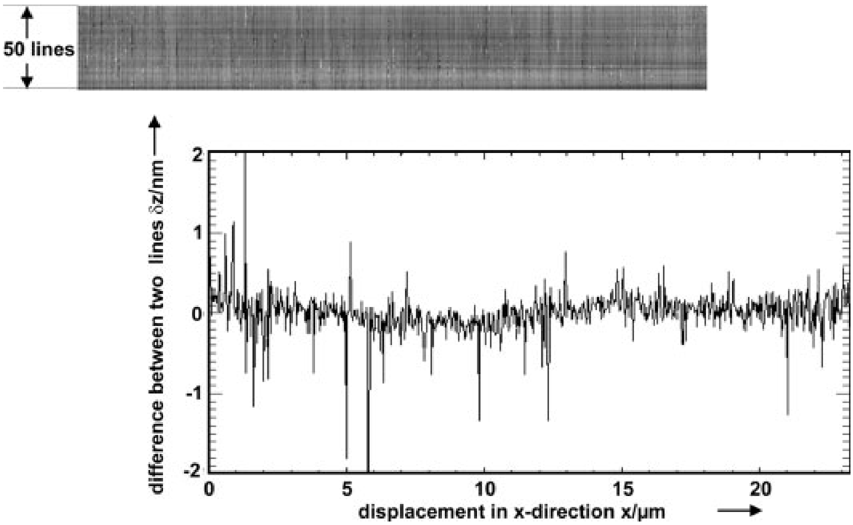

Fig. 13.6 STM with capacitive transducers [3], which was located in the PTB clean room center. (a) Image obtained in the flat area. No significant changes of the profile occur within these 50 scans. (b) The difference between two successive measured lines is displayed. The difference is clearly smaller than 0.5 nm (except for some spikes).

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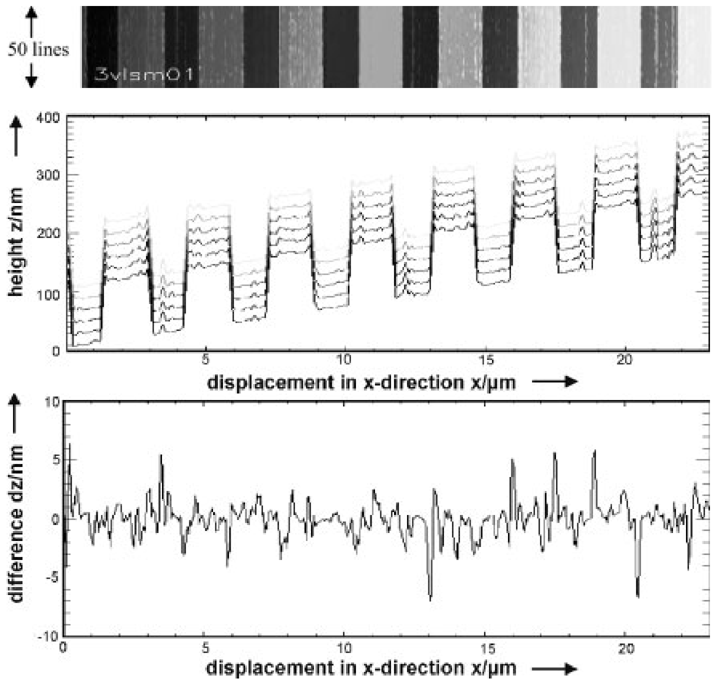

Fig. 13.7 (a) Repeatability of the STM discussed while scanning a structured region. (b) Profile lines taken from the data in (a). (c) Differences between two successive profile lines reveal a maximum deviation of 5 nm at the edges of the lines.

show that this happens at the transitions upward or downward. This behavior can be explained by small changes of the tip positions and by the sharp edges of the structure; as the tip and also the sample step edges are not ideal, e. g., as there is some roughness, small changes in the relative position between the tip and the sample lead to a change of the points at which the interaction happens.

13.3.4 Tip Shape

The SPM image can basically be understood as a convolution or a dilation of the actual surface features with the shape of the tip used for probing the sample [4]. To characterize the tip shape, various methods implying different evaluation algorithms can be applied [5].

Direct tip imaging methods allow us to read the tip shape from the SPM image [6]. Suitable specimens are upright tips or cylindrical features with (nearly) vertical edges. Straight edges allow us to deduce the cross-section of the tip body in one direction. The very first nanometers of a probe may also be characterized by gold particles (typical diameter: 5–30 nm) on a flat substrate.

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Test samples with sharp edges of random orientation such as TipCheck and NioProbe [7] may be applied for an indirect tip shape determination: The so-called blind reconstruction method [8] is an algorithm that calculates the "sharpness" of the tip necessary to obtain the actual image. The most critical aspect of this method is how to eliminate noise prior to the tip reconstruction calculation [9].

Apart from tip characterization in the SPM itself, scanning electron microscopy is usually a good means to study the probe.

13.4
Calibration of the Scanner Axes

13.4.1
Lateral Calibration

The lateral calibration of B- and C-type SPM systems is usually performed by a set of measurements at appropriate physical standards that should be chosen according to the actual needs (Figure 13.8). Such lateral standards with regular periodic structures of well-known dimension in one [10, 11] or two lateral directions [12] are meant to determine individual correction factors for the x- and y-axes. To reduce the influence of z-movements, the height of the structures should be as small as possible.

Before a decision is taken that which standards are actually to be used for the calibration, not only the pitch values including their uncertainties, the material

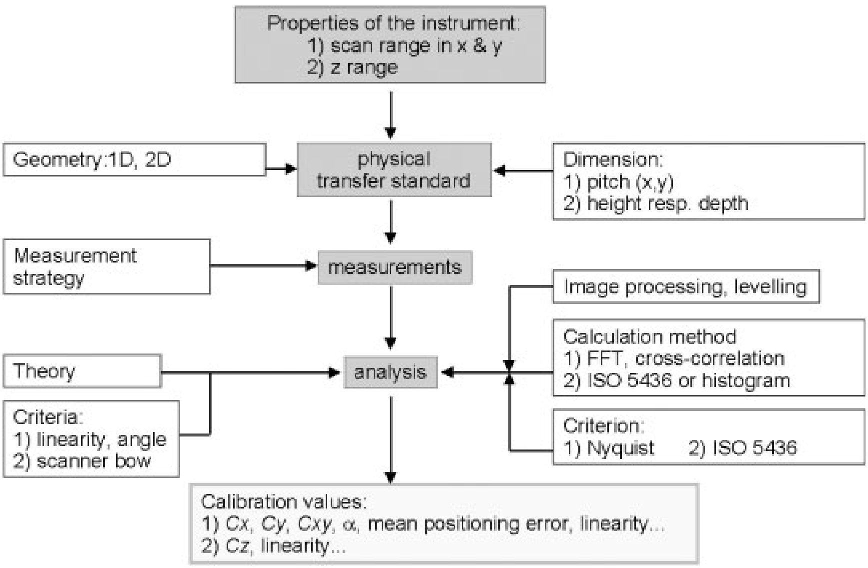

Fig. 13.8 Calibration scheme for (1) lateral calibration and (2) vertical calibration [13].

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composition, the structures' geometry respectively arrangement should be considered, but also the practicability of the standards, e. g., whether the size of the standard is suited for mounting and whether the standard allows sufficiently easy positioning in the apparatus (e. g., by orientation marks that guide the user to the reference field) to make sure that the certified area is found.

Another very important criterion often neglected is the homogeneity of the structures: Irregularities within the reference field such as possible gradual variations of the pitch value or significant periodic or occasional deviations [14] from the mean pitch–e. g., due to stitching errors of the lithography tools used to generate the structures–are, moreover, often not to be found in the documentation delivered with the standard or even not yet known at all. Especially when aiming at a high-quality calibration of SPMs of category B, such "hidden" irregularities may unnecessarily lead to a wrong calibration.

With the present evolution of large-range metrology SPMs, it is to be hoped that more knowledge of nanostandard irregularities will be gathered in the next years and consequently be included in the product descriptions and calibration certificates of nanostandards.

Investigations of a structured sample in different orientations 0°, 90°, 180°, and 270° provide information on the rectangularity of the x–y scanner system. The analysis can relate the x- and y-axes of the scanner to the **a** and **b** vectors of the surface unit cell. This analysis can be done by instruments or additional commercial software (see Figure 13.9).

A linear transformation (Figure 13.10) of the scanned and uncorrected image into a corrected image can be defined by a set of correction parameters Cx, Cy for the x- respectively y-axis and Cxy for the coupling between these two axes. Such transformations are integrated in some software packages to allow the user to correct scanner properties easily. The observed image is considered as linear, as the residual nonlinearity of the image is much less than one pixel for the actual measurement conditions.

To obtain information about the rectangularity of the x–z or y–z plane it is necessary to carefully investigate samples with well-defined topographical structures with finite slope. The analysis is described by Garnaes [16].

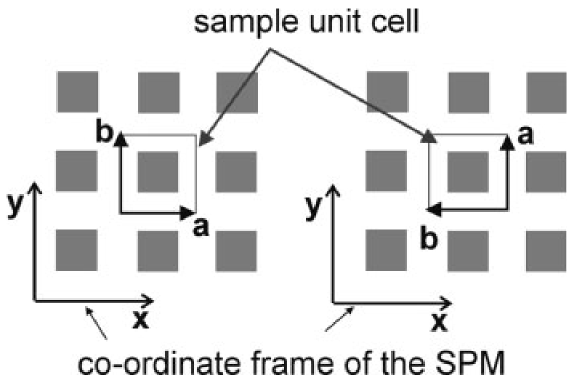

Fig. 13.9 Rectangularity within the x–y scanning plane can be obtained from at least two successive measurements with intermediate rotation of the test sample [12].

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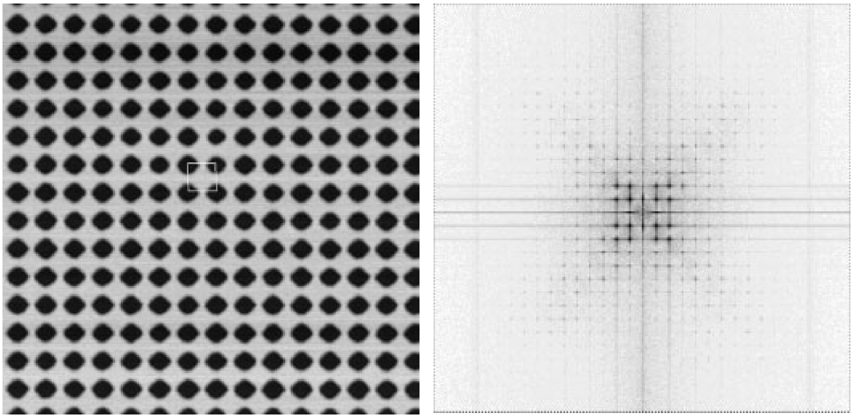

Fig. 13.10 SPM image of a 2D 1000 standard (14.67 μm × 14.48 μm) and the 2D FFT to determine the unit cell vectors and the angle ∡ (**a**,**b**) using SPIP [15].

Furthermore, lateral standards allow distortions in the measured image to be determined, i. e., positioning, straightness, squareness, and rotational errors of the scan system used (Figure 13.11). Depending on their origin, these errors are either temporary, permanent, or of a dynamic nature.

Temporary distortions might be caused by drifts due to temperature changes of the mechanical components, by initial piezo drifts or by the warm-up of electronic components. Such initial effects are to be avoided by making sure that the system is allowed a sufficient idle time to reach its equilibrium.

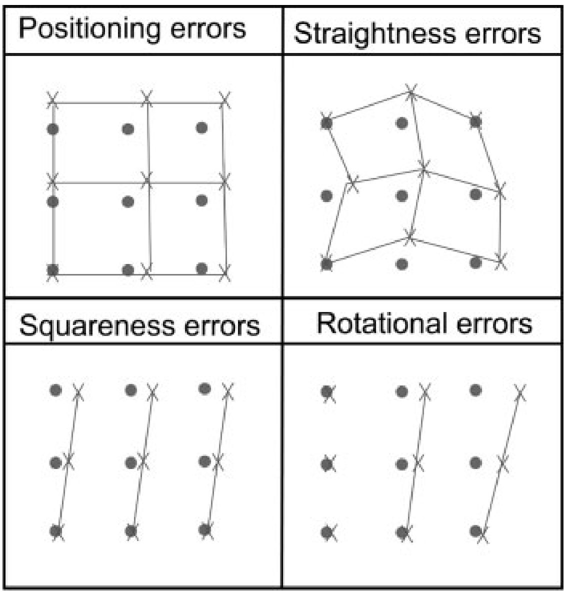

Fig. 13.11 Different types of image distortions. The small circles indicate the reference positions in the ideal two-dimensional grid; the crosses indicate the measured positions.

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Permanent distortions are mainly due to a nonideal design of the instrument such as a nonrectangular arrangement of the scan units, an insufficient mass balance inducing torque and mechanical stress, nonuniform guidance, etc. These effects need to be studied in order to check whether they can be taken into account systematically.

The notorious piezo hysteresis is, unless corrected by a closed-loop position control circuit, a permanent cause of image distortions, but strongly depends on the scan speed and scan range and is thereby of dynamic nature. Other dynamic error sources are a consequence of the frequency transmission functions of the various mechanical and electronic components, resulting in a low-pass-type retardation of the actual scan position to the desired position when the scan speed is set too high.

Some of the above-mentioned errors depend systematically on the actual scan parameters such as scan size and scan speed; consequently, calibration measurements usually need to be performed for several settings of these parameters.

By recording a series of calibration measurements at various scan sizes, a set of correction factors is acquired for these scan sizes that may serve as the basis to calculate a scan-size-dependent correction function. With the help of this function, images of any size may be corrected by an appropriate correction factor. It has to be taken into account, however, that such a correction function is usually based on only a few data points (usually 5–10) and therefore not very exact.

In many cases it proves reasonable to also take the dynamic behavior of the scan system into account when performing a lateral calibration. Moreover, the correction factors respectively image correction functions, which then need to be applied, often turn out to be more influenced by the dynamic behavior of the scan system than by the selection of the scan range.

Even in the case of a closed-loop position-controlled scanstage, scan-speed-dependent distortions are distinguishable [17]. Such a systematic influence of the scan speed allows one to calculate individual mathematic functions to correct the image distortions. Furthermore, depending on the actual system used, it is anyway often advisable to determine lateral correction factors for different scan speeds.

Further boundary conditions to be considered are the digital resolution (bit depth), electronic noise, and the number of pixels in an SPM image.

13.4.2 Calibration of the Vertical Axis

In many respects the calibration procedure for the vertical axis (z-axis) is comparable to that for the lateral axes. The calibration effort is therefore quite similar.

Like the lateral axes, also the movements along the vertical axis might show a nonlinear behavior and be affected by distortions. Moreover, as the vertical movements are often much faster, in order to follow the object's topography, than in the lateral scan directions, dynamic properties of the z-axis play an even larger role than in the case of the lateral axes. Consequently, much attention is to be

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paid to them. Furthermore, the parameters of the feedback control system (e. g., proportional factor P and integration time constant I) directly influence the dynamic behavior of the SPM in the z-direction and need to be adjusted carefully for the actual measurement, e. g., by monitoring the error signal and subsequently minimizing it.

Depending on the design of the instrument and the accuracy achievable respectively desirable, various strategies might be followed to characterize the z-axis. Two of them will be discussed in the following: direct laser-interferometric calibration and calibration by measurements at a set of transfer step-height standards.

Using Laser Interferometers

The elongation of the z-piezo can, at least in principle, be directly measured by laser-interferometry: In the case when the probe is moved in the z-direction, a small mirror is attached to the probe holder that serves as reflector for the laser beam. A deflection prism (or mirror) on the sample holder directs the laser beam from the vertical to the lateral direction and thus back to the interferometer. In the case when the sample is scanned in the z-direction, the mirror is placed on the sample holder and a small deflection prism is attached to the probe holder, respectively [18–20].

With this auxiliary measurement setup, the voltage externally applied to the z-actuator and the interferometer signal can be recorded simultaneously. In case the z-position is monitored by a sensor, this signal is recorded parallel to the interferometer reading. The particular advantage of this method is that it is ultimately versatile because the SPM user is – by choosing the shape of the voltage curve – free to simulate any "artificial topography" to study the system's response. Furthermore, a more direct traceability to the meter definition is achieved by using a possibly stabilized laser of calibrated wavelength.

For instance, the voltage might be increased from the minimum to the maximum allowed level in equal steps; this method allows us to determine the nonlinearity along the z scan range in a similar way as a pitch standard across the lateral scan directions. Based on these data, a correction function can be calculated. Secondly, the time constant of the stage's response can be read directly from the interferometer signal at any voltage step.

The dynamics of the z scan system can easily be assessed by varying the duration of the voltage steps. As piezo creep and hysteresis might be significant, it might be advisable in some cases to calculate a set of correction functions for the z-values for different speeds of the voltage ramp depending on the particular SPM system.

It has to be noted, however, that this method works "without load," i. e., the system is not in active distance control feedback. Switching to feedback control will consequently change the time constants of the system's response depending on the P and I parameter settings of the PI feedback system.

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Using Transfer Standards

While not every SPM user has a laser-interferometer at his or her disposal and the geometry of many SPM systems does not allow the integration of an optical interferometer, physical transfer standards are often used. They are easier to handle and can be used in any SPM system. The disadvantage is, however, that the traceability chain is longer than for the laser-interferometric method: The uncertainty of the transfer standard adds to the overall uncertainty of the SPM calibration. The advantage is that the transfer standards are measured under the same conditions as in the experiments later (i. e., in active probe feedback).

In order to take the nonlinearity of the z scan system into account, a set of standards of different step height should be used. In a first calibration approach, the z offset may be adjusted, if possible, in such a way that the z actuator operates in the middle of its range, i. e., symmetrically around 50 % of its maximal elongation during each of the calibration measurements (Figure 13.12). Provided that all future measurements of the actual measurement objects will adhere to this rule, satisfactory results will most likely be achieved in this way. If this is not possible, the same calibration procedure may be repeated around other mean elongations, e. g., around the 10 %, 30 %, 70 %, and 90 % threshold of maximum piezo span as depicted in Figure 13.12.

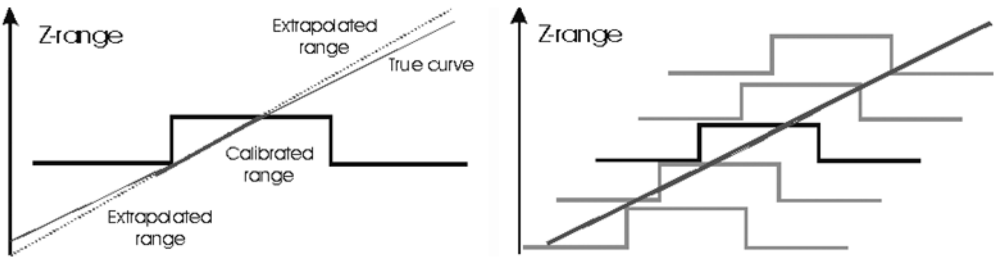

Fig. 13.12 Calibration of the z-axis using transfer standards. (a) single-point and (b) multiple-point calibration.

Evaluation of Step Height

Two methods of data analysis have proven to be suited for the determination of the correction factor for the z-axis: the histogram method (Figure 13.13) and ISO 5436 (Figure 13.14). Both methods yield similar results that usually agree within the uncertainty limits.

A precondition for the successful application of the histogram method is that the sample tilt has already been subtracted. In many cases a first-order plane correction will not be sufficient. It is therefore advisable to start with a measurement at a flatness standard prior to the height measurements in order to be able to subtract the flatness error from the step-height images. The histogram method itself comprises a plot of the height value distribution, the identification of the two peaks corresponding to the two height levels, the determination of the center

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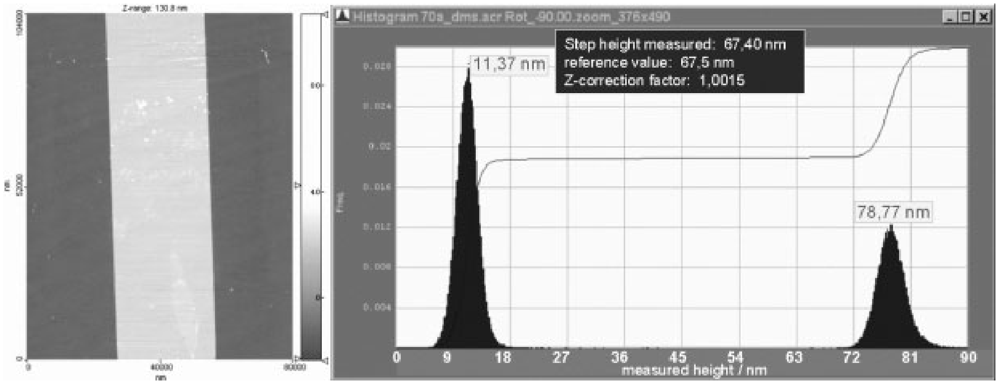

Fig. 13.13 Measurement (80 μm × 104 μm) at a step-height standard and corresponding histogram plot.

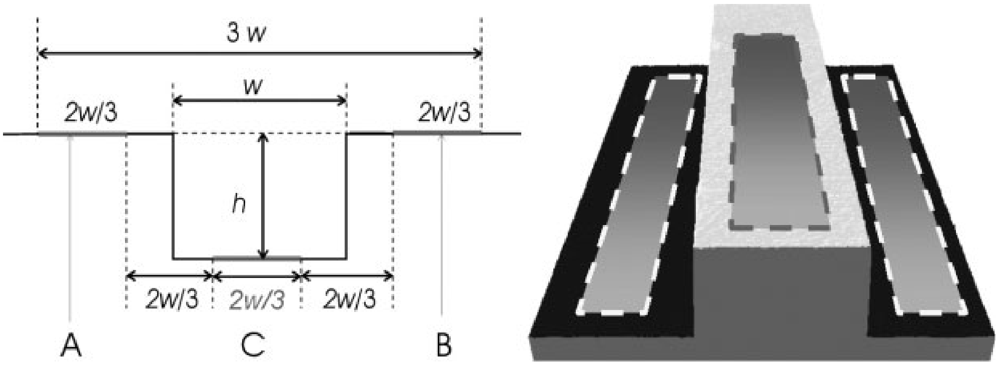

Fig. 13.14 Illustration of step-height determination according to ISO 5436: The central part of a one-dimensional structure (here 2/3 of its width, C) and two sections A & B of the same size as C and equally spaced from the edges are taken into account, whereas the edge region is left out of consideration.

of gravity in each of these two peaks, and calculating their difference (Figure 13.13).

The principle of ISO 5436 is illustrated in Figure 13.14: The central part of a positive or negative bar is chosen to fit a line through these values. As reference, two equally sized sections on both sides of the 1D structure are selected with equal spacing to the edges of the bar, and a line is also fitted through these values. The height value of the bar is defined as the normal distance of both fit lines. ISO 5436 may be applied either for a single scan line only or subsequently for several lines respectively the whole scan image–with the latter allowing us to determine variations in the height values measured along the 1D structures.

In this way, the edges themselves are not taken into consideration, which is a practical advantage of ISO 5436: Apart from the fact that edges often suffer from contamination and corner rounding, feedback errors are usually much larger at

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edge transitions than on the flat areas. Furthermore, optical techniques for the cantilever bending/oscillation detection such as the commonly used beam deflection or the focus sensor often not only detect the measuring beam reflected from the probe as desired, but also unwanted reflections from the sample surface that consequently spoil the signal used for the feedback and thereby height measurement; ISO 5436 reduces the influence of this parasite signal as such effects occur especially at edges.

A key problem in using transfer step-height standards is how to deal with contaminations—no matter whether the histogram method or ISO 5436 is applied. Unfortunately, especially smaller dust and dirt particles can often not be removed from the standards completely.

13.5 Uncertainty of Measurements

The uncertainty has to be calculated for the measurements following the rules of the *Guide to the Expression of Uncertainty in Measurement* (GUM) [21]. For this, the properties of the instrument and of the experimental setup have to be taken into account as well as the properties of the standard and the sample. Here we will give an example for the calibration of a step height on an unknown sample by a single measurement. Repetition of this measurement would reduce uncertainty contributions that are based on random errors, but not reduce systematic deviations.

The instrument has been calibrated by a given certified standard of height h_c together with the expanded uncertainty U_{95} using a coverage factor of $k = 2$. The standard used should have a height $h_{c,}$ which is close to that of the unknown sample h_x. For the calculation of the uncertainty, a model has to be set up. In our case the model is given by

$$h_x = \frac{h_c}{h_{cm}} h_{xm} + \sum \delta h_i$$

where h_x is the value of the step height of the unknown sample, h_c is the value of the step height of the standard given in the certificate, h_{cm} is the measured value on the standard, h_{xm} is the measured value on the unknown sample, and δh_i are other contributions to the measurement uncertainty.

The other contributions are related to errors due to stability and quality of the standard, remaining nonlinearity of the scanner, cross-talk between the x–y and z-axis, roughness and homogeneity of the standard and the sample, type of evaluation method, drift effects in the scanner during the measurement, and so on. Below heights of 1 μm a correction of the step height due to deviations from 20°C is negligible for standard materials.

The standard uncertainty $u_c(h_x)$ calculated from the above model is given by

$$u_c^2(h_x) = u^2(h_c) + u^2(h_{cm}) + u^2(h_{xm}) + u^2(\delta h_{\text{non-lin}}) + u^2(\delta h_{\text{cross-talk}}) + u^2(\delta h_{\text{eva.}}) + \cdots$$

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Here we considered a case in which $h_c \sim h_{cm} \sim h_{xm}$. The expanded uncertainty U_{95} has to be calculated from $u_c(h_x)$ using a coverage factor of $k = 2$. For more detailed calculations a greater knowledge of the instrument is necessary.

For more information about uncertainty budget calculation for pitch and step height, see [10, 11, 18–20].

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